

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Extended Reality Technologies [NPT_4135]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is the fundamental difference between Mixed Reality (MR) and Augmented Reality (AR)? (2)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) What is the key characteristic of Spatial Audio that differentiates it from normal stereo sound? (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Discuss its four key components of Virtual Reality. Provide real-world examples to illustrate each component. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) What is the primary function of a Three-Dimensional Position Tracker? (2)
- A) Course Outcome: "CO3" / Cognitive Level: "Understand"
- B) Categorize is the key principle behind the "Lighthouse" tracking approach used by devices like the Valve Index? (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Illustrate the problem of "drift" in IMU-based tracking and describe how a typical VR system compensates for it using a technique like sensor fusion. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Briefly explain what an Animation State Machine is used for. (2)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Briefly explain how a multiple-camera infrared tracking system (like the HTC Vive) works. (3)
- Course Outcome: "CO4" / Cognitive Level: "Understand"
- C) Describe the process of setting up a simple AR experience using the Vuforia SDK. Your answer should include at least three key steps, from target creation to content display. (5)
- Course Outcome: "CO4" / Cognitive Level: "Application"

4) List out the primary cause of Simulator Sickness in VR users? (2)

A) Course Outcome: "CO2" / Cognitive Level: "Analysis"

B) How does a "Lighthouse" tracking system differ from a traditional infrared tracking system that uses cameras? (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

C) Differentiate between Rigidbody and Collider in Unity. Why are both necessary? List the different types of Colliders? (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) What is the primary purpose of using Inverse Kinematics in avatar tracking? (2)

A) Course Outcome: "CO3" / Cognitive Level: "Application"

B) Write the key principle behind the "Lighthouse" tracking approach? (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

C) Write the main challenges associated with Outdoor Tracking for XR experiences? Discuss at least two specific challenges. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	NPT_4135:Extended Reality Technologies

#	Course Outcome	Description	Marks
1	2	CO 2	15
2	3	CO 3	15
3	4	CO 4	20
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	NPT_4135:Extended Reality Technologies

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	7(14%)
3	Application	9	36(72%)
4	Analysis	3	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)